Tribhuvan University

Faculty of Humanities and Social Science

Shop Sphere

(E-Commerce System)

A PROJECT REPORT

Submitted To

Department of Computer Application

Shahid Smarak College

In partial fulfillment of the requirements for the Bachelor in Compute Application

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Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by **Bibash Bayalkoti** and **Nikesh Shrestha** entitled **"Shop Sphere"** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

_____

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LETTER OF APPROVAL

This is to clarify that this project prepared by **Bibash Bayalkoti** and **Nikesh Shrestha** entitled **"Shop Sphere"** in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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Abstract

Shop Sphere is an eCommerce platform developed to provide a modern, efficient, and user-friendly online shopping experience. The main objective of this project is to develop a complete eCommerce system that allows users to browse products, add items to a cart, and place orders smoothly while ensuring secure authentication and data management. The system also implements an intelligent product recommendation feature to personalize the shopping experience and improve user satisfaction. Additionally, the platform enables administrators to efficiently manage products, users, and orders through a dedicated admin panel. By combining personalization, usability, and efficient management tools, Shop Sphere aims to enhance customer engagement and increase overall sales performance.

Acknowledgement

We would like to express our sincere gratitude to everyone who supported us during the development of the Shop Sphere project. We are especially thankful to our mentors and teachers for their valuable guidance, suggestions, and encouragement throughout the project. We also appreciate the support and motivation from our friends and well-wishers who contributed directly or indirectly to the successful completion of this project. Finally, we are deeply grateful to our families for their continuous support and encouragement, which helped us complete this project successfully as a team.

Nikesh Shrestha

Bibash Bayalkoti

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Chapter 1: Introduction

1.1 Introduction

E-commerce has transformed the way people shop by providing fast, convenient, and accessible online marketplaces. With the growth of internet usage and digital platforms, customers now expect personalized shopping experiences. A recommendation system enhances user satisfaction by suggesting products based on user behavior, preferences, and trends.

"Shop Sphere" is a web-based e-commerce platform focused on clothing brands, enabling individuals or organizations to browse, select, and purchase a wide range of apparel items over the internet. It will empower users to explore trending clothing from various brands, with personalized recommendations powered by a machine learning algorithm. This project aims to bridge the gap between fashion enthusiasts and quality clothing suppliers, reducing the need for physical store visits and minimizing business overhead costs.

Beyond basic online shopping functionalities, Shop Sphere's core feature will be an integrated product recommendation system. This system will analyze user behavior and product characteristics to suggest relevant items, personalizing the shopping experience and encouraging further exploration.

The main purpose of this system is to deliver high-quality clothing products and services to customers anytime, anywhere, via secure online transactions. Additionally, the recommendation algorithm will enhance user engagement by suggesting relevant items, improving the overall shopping experience.

1.2 Problem Statement

Many existing online shopping systems in developing countries face the following problems:

- Users find it difficult to discover relevant products quickly.
- Lack of personalization in product suggestions.
- Manual product searching consumes more time.
- Poor user experience due to non-intelligent systems.

1.3 Objectives

The main objectives of the Shop Sphere are as follows:

- To develop an ecommerce platform.
- To implement an intelligent product recommendation system.
- To provide a user-friendly shopping interface.
- To increase sales and user satisfaction through personalization.

- To allow admins to efficiently manage products, users, and orders.

1.4 Scope and Limitation

Scopes

- To develop a complete and functional eCommerce platform.
- To provide secure user registration and login functionality.
- To allow users to browse products and view detailed product information.
- To enable users to add products to a shopping cart and place orders easily.
- To implement an intelligent product recommendation system for personalized shopping.
- To provide a user-friendly and responsive interface.
- To allow administrators to manage products, users, and orders efficiently.
- To build a scalable system that can be enhanced with advanced features in the future.

Limitations

- The recommendation system may be basic and not use advanced machine learning algorithms.
- Online payment gateway integration may be limited or not fully implemented.
- The system may not support multi-vendor functionality.
- Real-time inventory management may be limited.
- Advanced sales analytics and reporting features may not be fully developed.
- The platform may require further optimization to handle very high traffic.
- Mobile application support may not be included.

Chapter 2: Background Study and Literature Review

2.1 Background Study

The rapid advancement of internet technology and digital communication has significantly transformed the way businesses operate and how consumers purchase products. eCommerce platforms have emerged as powerful tools that enable customers to browse, compare, and purchase products online without geographical limitations. Popular platforms such as Amazon, eBay, and Alibaba have demonstrated how online marketplaces can successfully replace or complement traditional physical stores. These platforms provide user-friendly interfaces, secure payment systems, product management features, and personalized recommendation systems to enhance user engagement and increase sales.

Modern eCommerce systems focus not only on selling products but also on improving customer experience through personalization and intelligent recommendations. Recommendation systems analyze user behavior, preferences, and purchase history to suggest relevant products, thereby increasing customer satisfaction and business profitability. Additionally, administrative dashboards play a crucial role in managing products, users, and orders efficiently. Inspired by these existing platforms and technological advancements, Shop Sphere is developed as an eCommerce solution that integrates essential online shopping functionalities with intelligent product recommendations and efficient management tools to create a smooth and engaging shopping experience.

Study of existing system

Daraz Nepal

Daraz Nepal is one of the largest eCommerce platforms in Nepal, offering a wide variety of products such as electronics, fashion, home appliances, and groceries. It provides features like online payment, cash on delivery, order tracking, and customer reviews. Daraz operates as a multi-vendor marketplace, connecting sellers and buyers across the country.

SastoDeal

SastoDeal is a Nepali online shopping platform that offers products including electronics, lifestyle items, and household goods. It focuses on providing affordable products with home delivery and online payment options. The platform aims to deliver a convenient shopping experience to customers within Nepal.

Hamrobazar

Hamrobazar is a popular online classified marketplace in Nepal where users can buy and sell new or used products directly. It mainly allows individuals to post advertisements for items such as vehicles, property, electronics, and jobs. Unlike full eCommerce platforms, it connects buyers and sellers without handling complete transaction management.

2.2 Literature Review

The advancement of technology has significantly impacted the e-commerce sector, particularly in fashion retail. The integration of recommendation systems has transformed how consumers discover and purchase products, leading to improved personalization and business outcomes. This literature review examines key aspects: user interface/experience, consumer behavior in online fashion shopping, and the role of recommendation algorithms.

A well-designed user interface and experience are essential for e-commerce success, especially in visually driven domains like clothing. According to Pallavi K N and Dhanya Sharanya Shree Rin [1], aesthetically pleasing designs with engaging visuals, intuitive layouts, and consistent branding create positive first impressions and foster user loyalty.

Consumer behavior in online clothing shopping is influenced by convenience, personalization, price comparison, and reviews. Megan Wenzl [2] highlights that trends such as mobile accessibility, seamless payments, and AI-driven suggestions are critical for platform success. Recommendation systems, using techniques like collaborative filtering or content-based filtering, significantly enhance product discovery. Sophia Fennessy [3] notes that these systems increase sales, customer engagement, and loyalty by providing relevant suggestions, reducing choice overload, and improving SEO through user-generated interactions.

Chapter 3: System Analysis and Design

3.1 System Analysis

The development of the proposed e-commerce platform Shop Sphere will follow the Agile Software Development Methodology. Agile is an iterative and incremental approach that focuses on flexibility, continuous improvement, and frequent feedback from users and stakeholders. This methodology is well-suited for web-based applications where requirements may evolve over time. This project will adopt the Agile methodology due to its flexibility for iterative development, especially with the integration of a recommendation algorithm that may require refinements based on testing. The technology stack is well-suited for modern web applications, with clear requirements but room for enhancements.

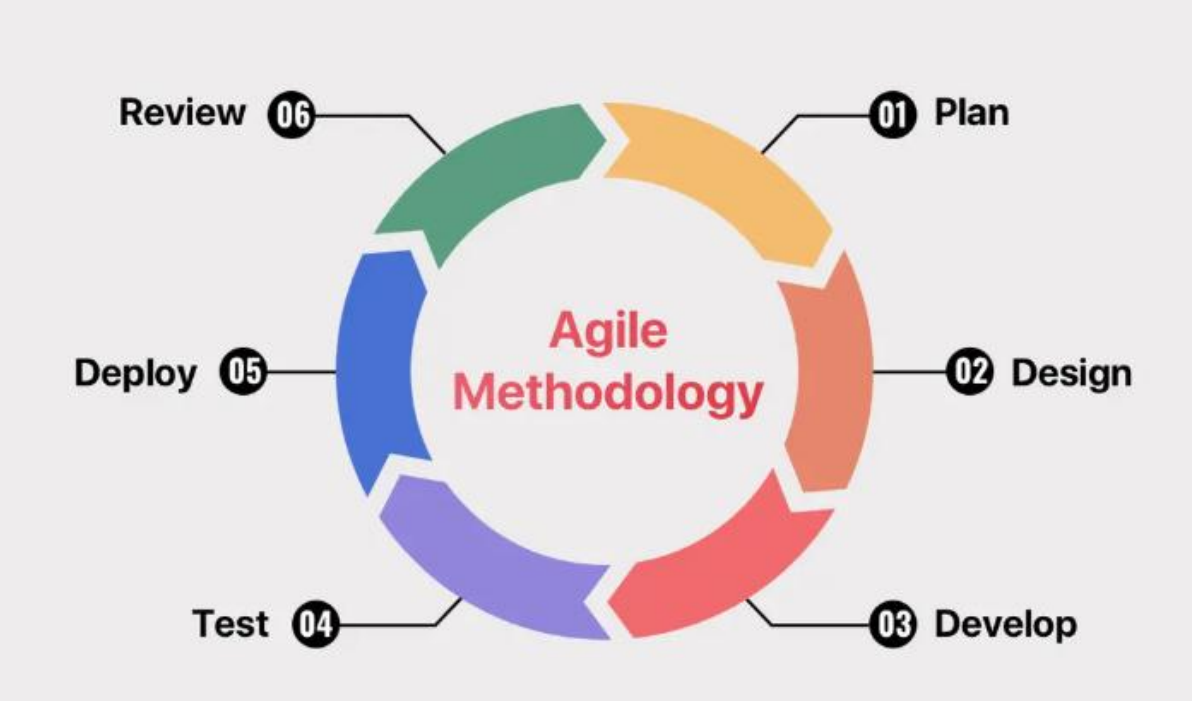

Figure 1 Agile Methodology

3.1.1 Requirement Analysis

Requirement analysis is the process of identifying and understanding the needs of users and the system before development. For Shop Sphere, it involves determining the essential features such as user registration and login, product browsing, shopping cart management, order placement, and an intelligent product recommendation system. It also includes administrative requirements like managing products, users, and orders efficiently. In addition, the system must ensure security, good performance, user-friendly design, and scalability to provide a smooth and reliable online shopping experience.

i. Functional Requirements

- The system shall provide secure user authentication with role-based access for Admin, Seller, and Customer.
- The system shall allow sellers to manage products by adding, updating, viewing, and deleting product information including price, description, category, and stock details.
- The system shall allow customers to browse, search, and view product details, add items to a shopping cart, modify cart contents, and place orders successfully.
- The system shall integrate online payment functionality using eSewa to enable customers to complete transactions securely.
- The system shall verify payment status and store transaction details after successful payment processing.
- The system shall generate order confirmation for customers and notify sellers after successful payment.
- The system shall provide an admin panel that enables the admin to manage users, monitor products, track orders, and oversee payment records efficiently.
- The system shall implement a product recommendation feature to personalize the shopping experience and enhance user satisfaction.

Use case diagram

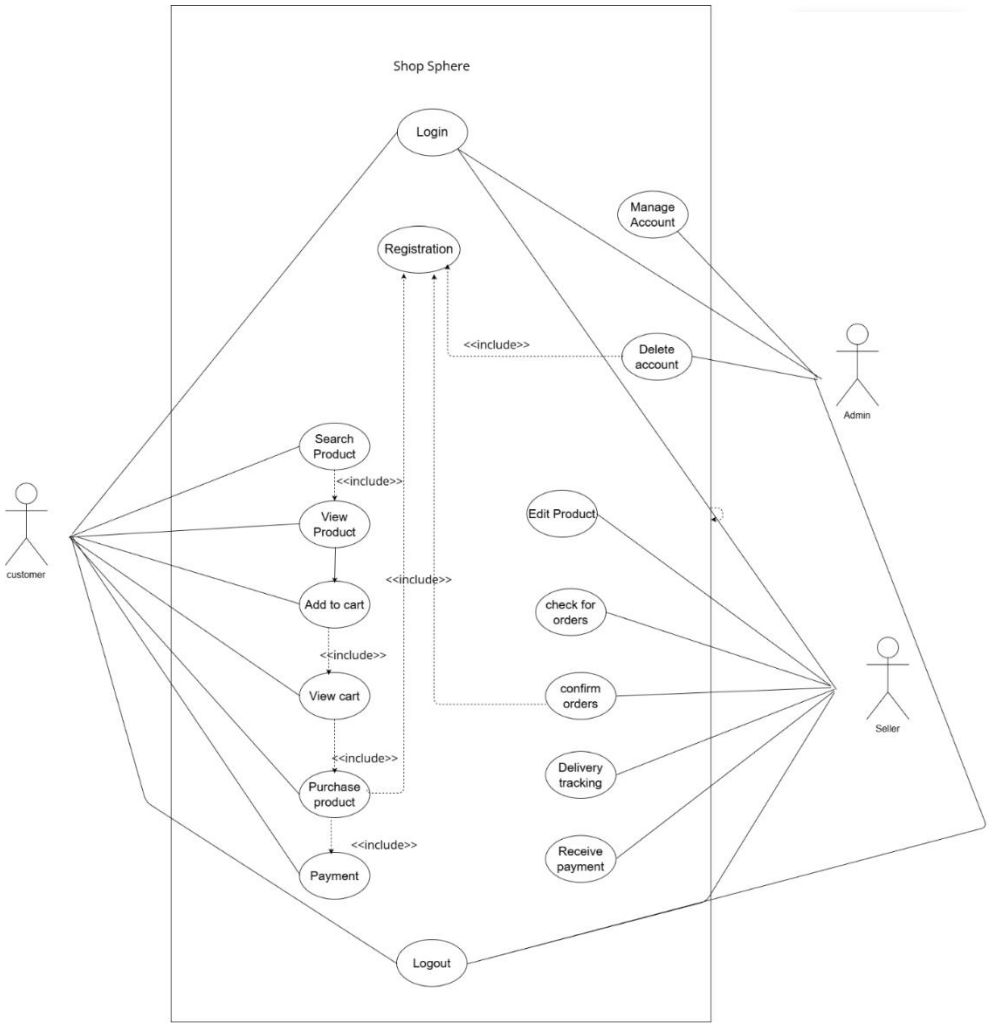

Figure 2 Use Case Diagram

ii. Non-Functional Requirements

- The system shall provide fast response time for product browsing, cart operations, and order processing, even when multiple users access the platform simultaneously.
- The system shall ensure secure authentication, encrypted password storage, and secure online payment processing through eSewa integration to protect user and transaction data.
- The system shall provide a simple, user-friendly, and responsive interface that allows users to navigate and complete purchases easily.
- The system shall operate consistently without data loss and ensure accurate storage of user, product, order, and payment information.
- The platform shall be designed to handle future growth in the number of users, products, and transactions.
- The system shall be accessible online at all times except during scheduled maintenance.
- The system shall be structured in a modular way so that updates, bug fixes, and feature enhancements can be implemented easily.

3.1.2 Feasibility Analysis

Technical Feasibility

The development of Shop Sphere is technically feasible as it utilizes reliable and widely used technologies. The frontend of the system is developed using HTML, CSS, and JavaScript to create an interactive and responsive user interface. The backend is implemented using Python, which ensures efficient server-side processing and business logic handling. PostgreSQL is used as the database management system to securely store and manage user, product, order, and payment data. These technologies are well-supported, scalable, and suitable for building a secure and high-performance eCommerce platform. Additionally, integration with eSewa for online payment further strengthens the technical capability of the system.

Operational Feasibility

The proposed system is operationally feasible as it is designed to provide a simple, efficient, and user-friendly online shopping experience. Shop_Sphere addresses common challenges faced in traditional shopping and existing online platforms by allowing customers to browse products, receive personalized recommendations, and complete secure transactions easily. The system also enables sellers to manage products effectively and administrators to monitor overall operations. The platform is easy to operate, requires minimal training, and can be accessed from any location with internet connectivity, making it practical for real-world use.

Economic Feasibility

The project is economically feasible because it is developed using open-source and freely available technologies, which significantly reduces development costs. There is no

requirement for expensive software licenses or proprietary tools. The infrastructure requirements are minimal, and the system can be deployed on affordable hosting services. Therefore, the overall cost of development and implementation is low compared to the potential benefits of increased sales, improved customer satisfaction, and efficient business management.

3.1.3 Data Modelling

ER -Diagram

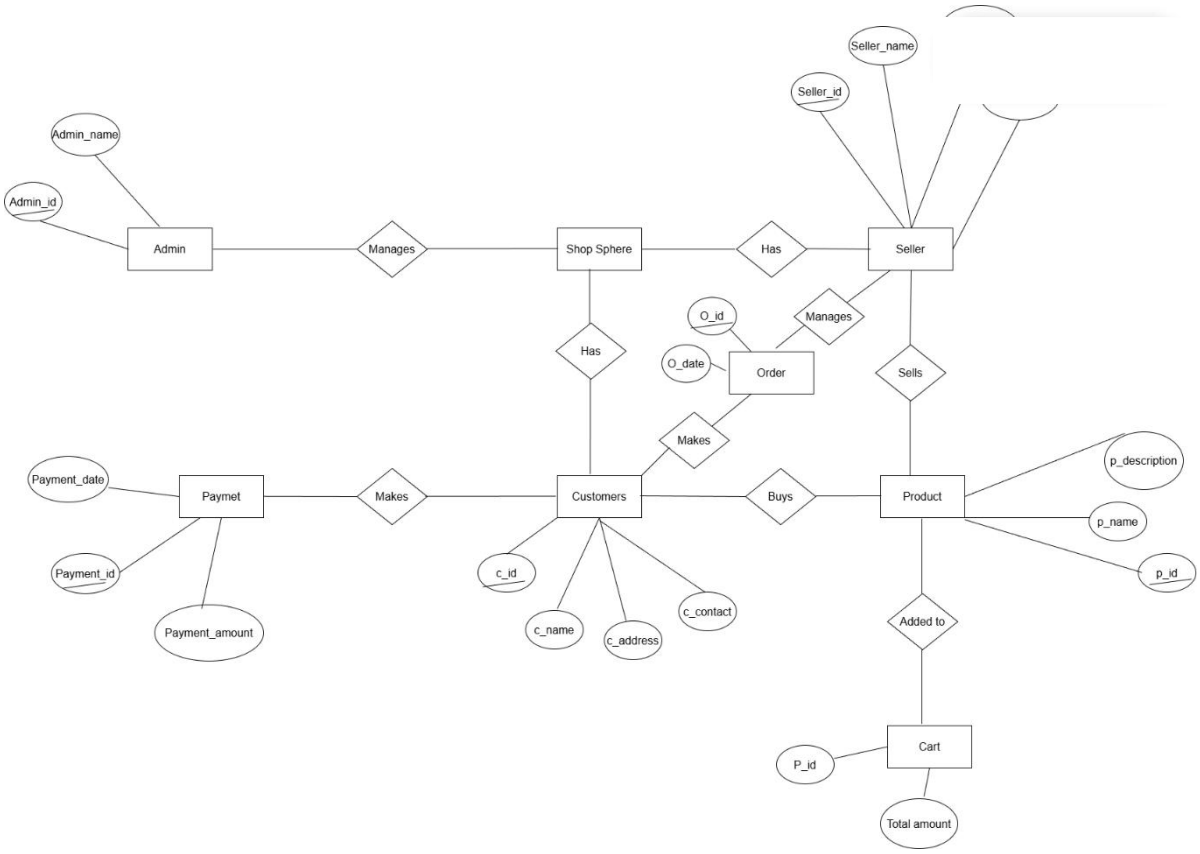

Figure 3 ER-Diagram

Flowchart

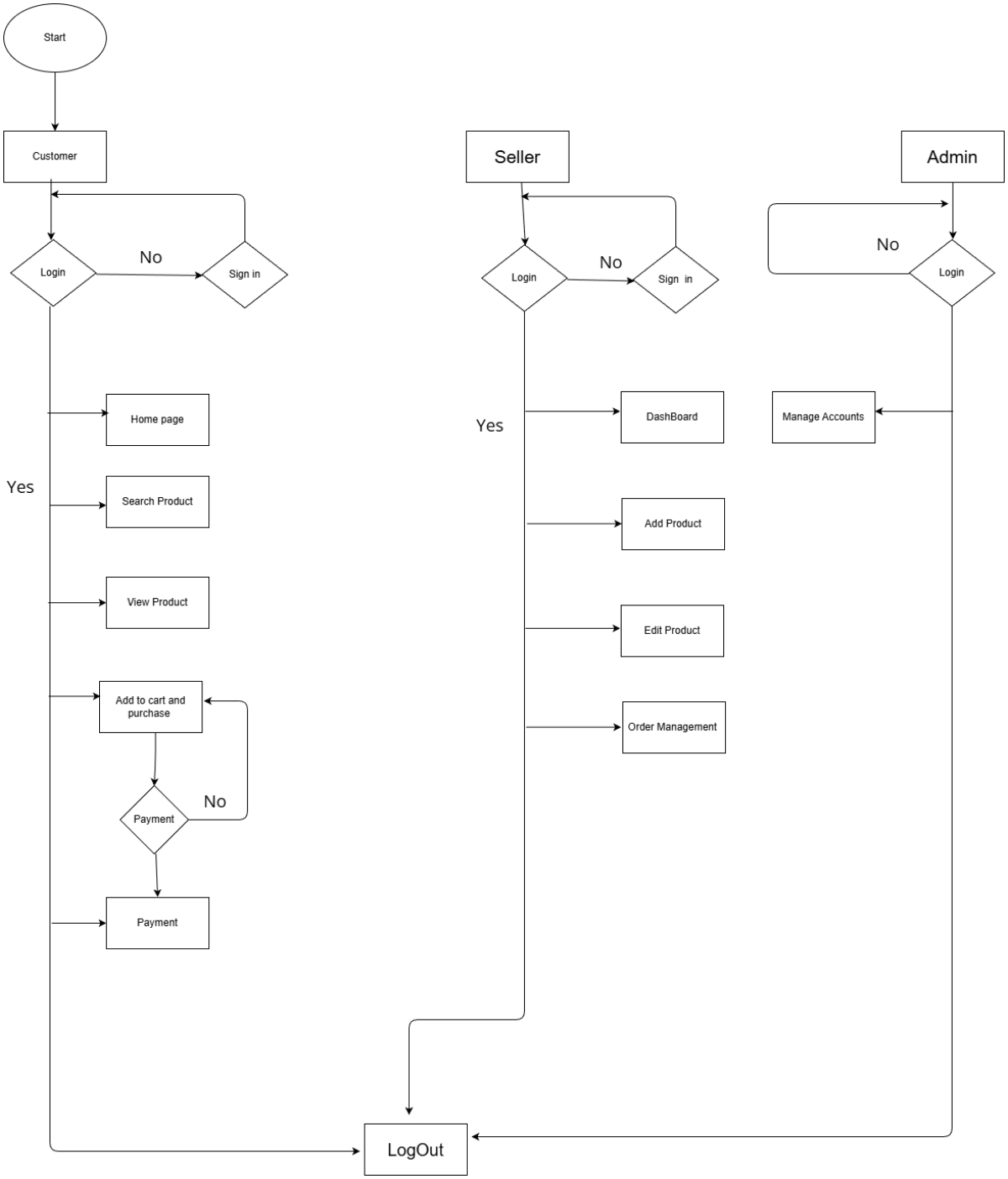

Figure 4 FlowChart

3.1.4 Process Modelling

Data Flow Diagram

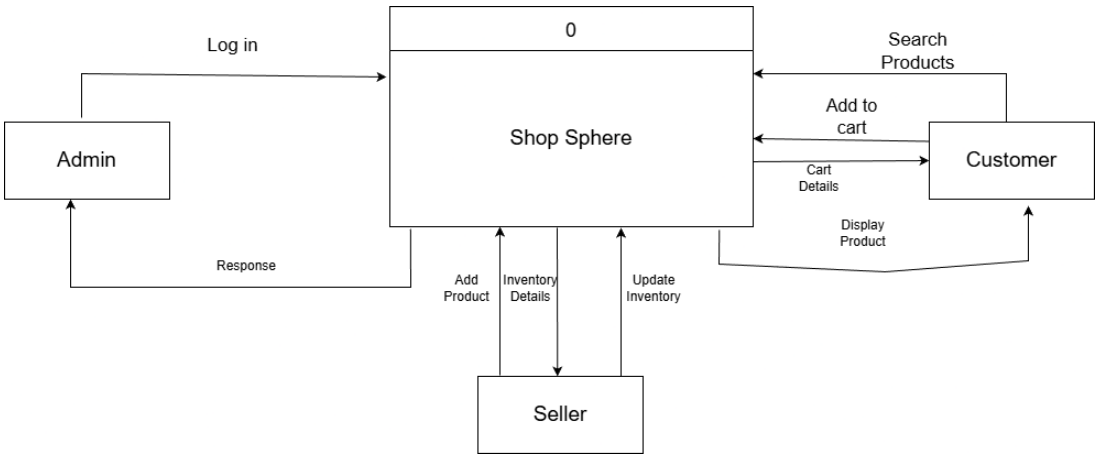

Figure 5 DFD Level 0

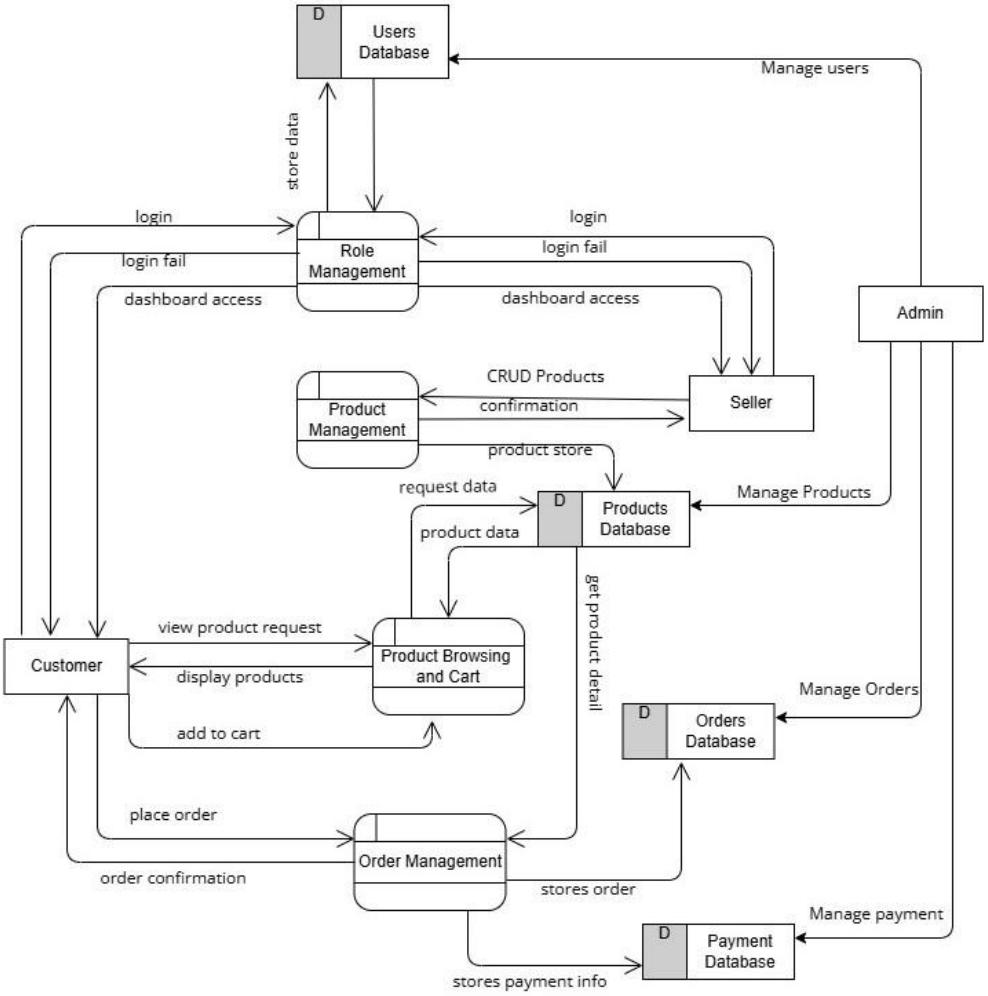

Figure 6 DFD Level 1

3.2 System design

3.2.1 Architectural Design

Client-Side (Frontend)

Purpose: The frontend is responsible for the User Interface (UI), allowing customers, sellers, and admins to interact with Shop_Sphere smoothly. It ensures a responsive, intuitive, and interactive experience for all users.

Technology Used:

- HTML, CSS – For structuring and styling the web pages.
- JavaScript – For dynamic elements, form validation, and enhancing interactivity.

Key Functionalities:

- User Authentication: Customers, sellers, and admins can register, log in, and log out securely.
- Product Browsing & Cart: Customers can browse, search, filter products, view details, and manage shopping cart items.
- Order Placement & Confirmation: Customers can place orders, make payments via eSewa, and receive confirmation notifications.
- Admin & Seller Panels: Sellers can manage products, and admins can manage users, products, orders, and payments through an intuitive dashboard.

Server-Side (Backend)

Purpose: The backend manages all the logic, processes, and communication between the frontend and the database. It ensures the system runs securely, efficiently, and accurately.

Technology Used:

- Django (Python) – Handles business logic, request processing, authentication, and integration with payment gateways.

Key Functionalities:

- User Authentication & Role Management: Ensures only authorized users (customers, sellers, admins) can access their respective features.
- Product & Order Management: Handles product CRUD operations, order processing, stock updates, and notifications.
- Payment Integration: Processes online payments securely using eSewa and updates transaction records.
- Recommendation System: Provides personalized product suggestions based on user behavior and purchase history.

Database (Storage)

Purpose: The database stores and manages all critical data related to users, products, orders, and payments.

Technology Used:

- PostgreSQL – A reliable relational database for structured and secure data storage.

Key Functionalities:

- Data Storage: Saves user profiles, product details, cart information, orders, payment records, and admin logs.
- Data Retrieval: Supports efficient queries for product browsing, order tracking, dashboards, and reporting.
- Security & Integrity: Ensures data consistency with constraints, transactions, access control, and backups.

3.3 Algorithm Details

Content Based Recommender System:

Recommendation systems are an important part of many digital platforms, like for suggesting movies on Netflix, recommending products on Amazon etc. Among the different types of recommendation approaches, Content-Based Recommender Systems focus on the characteristics of items and the preferences of users to generate personalized recommendations. It uses information about a user's past behavior and item features to recommend similar items.

A Content-Based Recommender System (CBRS) is a type of information filtering system that suggests items to a user by matching the attributes of the items with a profile of the user's preferences. Unlike collaborative filtering, which depends on the behavior of other users, this algorithm relies solely on the metadata (features) of the items and the historical interactions of a single user. It operates on the principle: "If a user liked an item with specific characteristics in the past, they will likely enjoy similar items in the future."

This can include explicit feedback such as ratings or even implicit feedback like clicks, views or time spent on content. Based on this data, the system generates a user profile, which is then used to find items that closely match the user's preferences. As time passes, the user continues to interact with the system which results in more accurate and relevant suggestions.

Theoretical Framework (The Algorithm):

The algorithm follows a three-step mathematical process to generate recommendations.

i. Item Profile Generation

Each item in the system is represented as a feature vector. For textual data, the TF-IDF (Term Frequency-Inverse Document Frequency) algorithm is typically used to convert text into a numerical vector.

- Term Frequency (TF): Measures how frequently a term occurs in a document.
- Inverse Document Frequency (IDF): Measures how important a term is across the entire corpus.

ii. User Profile Building

The system constructs a User Profile Vector based on the features of items the user has previously interacted with (rated, liked, or viewed). This vector represents the user's "weighted" interest in specific features.

iii. Similarity Computation (Cosine Similarity)

To determine the relevance of a new item to a user, the system calculates the similarity between the User Vector (u) and the Item Vector (i). The most widely used metric for this is Cosine Similarity, which measures the cosine of the angle between the two vectors.

Cosine Similarity Formula

$$\text{Similarity}(A, B) = \frac{A \cdot B}{\|A\| \times \|B\|}$$

Where:

- $A \cdot B$ = dot product of vectors
- $\|A\|$ = magnitude of vector A
- Result ranges from **0 to 1**
  - **1** → very similar
  - **0** → not similar

Algorithm Workflow

- Feature Extraction: Extract attributes (tags, categories, keywords) from all items in the database.
- Vectorization: Transform these attributes into a high-dimensional vector space using TF-IDF or Binary Encoding.
- User Preference Analysis: Aggregate the vectors of items the user has liked to create a summarized User Profile Vector.
- Filtering & Ranking: Calculate the similarity score between the User Profile Vector and all unrated items.
- Output: Sort the items by their similarity scores and recommend the top N items.

Chapter 4: Implementation and Testing

4.1 Implementation

Implementation is the process of translating the design into actual working code. In this phase, the system's components are developed, integrated, and tested to ensure they meet the specified requirements.

4.1.1 Tools Used

CASE Tools:

- Visual Studio Code – For coding, debugging, and version control.
- pgAdmin – For database management and query execution.

Programming Languages:

- Python (Django): Used for backend development, handling business logic, authentication, and server-side operations.
- HTML, CSS, JavaScript: Used for frontend development to create interactive and responsive web pages.

Database Platforms:

- PostgreSQL: Used as the relational database to store and manage all critical data such as users, products, orders, and payments.

Other Tools:

- eSewa API – Integrated for online payment processing.
- Git/GitHub – Version control and code management.

4.1.2 Implementation details of modules

User Authentication Module

This module handles user registration, login, logout, and role-based access control for Customers, Sellers, and Admins.

- **Classes Used:** Django's built-in User model, authenticate(), login(), logout(), and decorators for role-based access.
- **Implementation:**
  - Users register by providing required details (name, email, password).
  - Passwords are hashed using Django's authentication system.
  - Login verifies credentials and assigns appropriate dashboard access based on roles.
  - Admin, Seller, and Customer have restricted access to ensure security and proper functionality.

Product Management Module (Seller Panel)

This module allows sellers to manage their products efficiently using CRUD operations.

- **Classes Used:** Django Product model, ModelForm, views like add_product(), update_product(), delete_product(), and templates for product display.

- **Implementation:**
    - Sellers can add new products with details like name, description, price, category, stock, and images.
    - Existing products can be updated or deleted.
    - All product data is stored in PostgreSQL (Products Database) and dynamically displayed on the frontend.

Product Browsing & Cart Module (Customer Panel)

This module enables customers to browse products, add items to the cart, and prepare for checkout.

- **Classes Used:** Product views, session-based cart management, templates for product listing, and JavaScript for dynamic updates.
- **Implementation:**
    - Customers can search, filter, and view product details including price, stock, and description.
    - Adding items to the cart uses session management to store temporary cart data.
    - Customers can update quantity, remove items, or proceed to place an order.

Order Management Module

This module handles the entire order placement and payment process.

- **Classes Used:** Order and Payment models, Django views for order processing, eSewa payment integration API.
- **Implementation:**
    - Customers place orders from the cart.
    - Order details are stored in the Orders Database.
    - Payment is processed securely via eSewa, and transaction details are stored in Payments Database.
    - Upon successful payment, order confirmation is sent to the customer, and notification is sent to the seller.

Admin Control & Monitoring Module

This module allows admins to manage the platform efficiently.

- **Classes Used:** Admin views, User/Product/Order models, dashboard templates, and queries for monitoring.
- **Implementation:**
    - Admin can view and manage all users, products, and orders.
    - Admin can track payment status and generate reports.
    - The admin dashboard provides a consolidated view of the platform's operations, helping maintain smooth and secure system management.

Chapter 5: Conclusion and Future Recommendations

5.1 Conclusion

The Shop Sphere project successfully demonstrates the development of a modern eCommerce platform with user-friendly features, secure authentication, product management, shopping cart functionality, and online payment integration via eSewa. The system provides an intelligent product recommendation feature to enhance user experience and personalized shopping. The admin panel allows efficient management of users, products, orders, and payments, ensuring smooth operation of the platform. Overall, Shop_Sphere provides a reliable, scalable, and interactive solution for online shopping, addressing the needs of customers, sellers, and administrators effectively.

5.2 Lessons Learned / Outcome

- Gained practical experience in full-stack web development using Django, HTML, CSS, and JavaScript.
- Learned how to design and implement database structures in PostgreSQL for efficient data management.
- Understood the integration of online payment gateways (eSewa) in a web application.
- Improved knowledge of user authentication, role management, and secure transactions.
- Learned the importance of modular design for maintainability and scalability of a system.
- Developed skills in testing, debugging, and deploying a web-based project.
- Gained understanding of real-world eCommerce systems and how to implement personalized features like recommendation systems.

5.3 Future Recommendations

- **Multi-Vendor Support:** Allow multiple sellers to manage their own stores within the platform, increasing product variety and market reach.
- **Real-Time Inventory Management:** Integrate real-time stock updates to prevent over-selling and ensure accurate product availability.
- **Enhanced Payment Options:** Add multiple online payment gateways alongside eSewa, such as Khalti or IME Pay, to provide users with more flexibility.
- **Analytics & Reporting:** Implement advanced analytics and reporting dashboards for admins and sellers to monitor sales trends, customer behavior, and revenue growth.
- **Customer Support Integration:** Add live chat or AI-based support to assist customers quickly and improve user satisfaction.

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